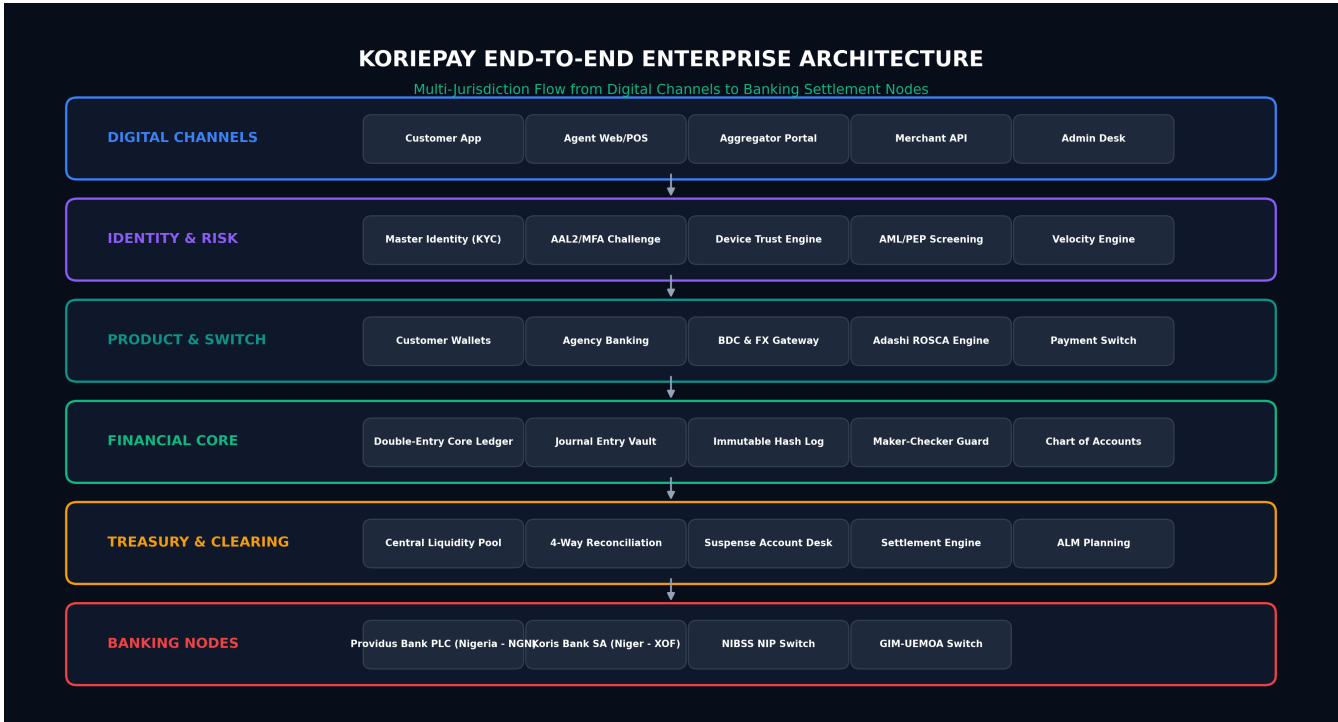


KORIEPAY

Confidential System, Financial Architecture, Compliance & Implementation Manual

A Unified Reference & Engineering Blueprint for Tier-1 Core Banking, Double-Entry General Ledger, Central Liquidity Pools, Adashi / ROSCA Orchestration, and Bilateral Financial Rails across Nigeria (NGN) and Niger Republic (XOF).

DOCUMENT CONTROL IDENTIFIERS	SPECIFICATION
Document Classification	STRICTLY CONFIDENTIAL // INTERNAL USE ONLY
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Mandatory Review Cycle	Quarterly or upon material system, infrastructural, or regulatory changes



MANDATORY REGULATORY & COMPLIANCE DISCLAIMER

REGULATORY FRAMEWORK & BOUNDARIES

This document is an internal technology, operational, financial architecture, and compliance-readiness manual prepared exclusively for authorized executives, engineers, auditors, and partner financial institutions of KoriePay. This document does not constitute formal legal advice, a direct banking licence, payment service bank licence, or standalone regulatory authorization. All financial operations in the Federal Republic of Nigeria operate in strict alignment with Central Bank of Nigeria (CBN) regulations, the Banks and Other Financial Institutions Act (BOFIA 2020), and commercial clearing nodes (Providus Bank PLC). All operations in the Republic of Niger operate within the West African Economic and Monetary Union (WAEMU / UEMOA) framework supervised by the Banque Centrale des États de l'Afrique de l'Ouest (BCEAO) in partnership with Koris Bank SA. Production deployment is strictly contingent upon verified regulatory filings, data localization compliance under the Nigeria Data Protection Act 2023 (NDPA), and active institutional maker-checker authorization protocols.

1.0 EXECUTIVE DOCUMENT PURPOSE & INTENDED AUDIENCE

This master technical and operational manual establishes a single authoritative baseline for KoriePay's enterprise architecture. It governs how capital flows across borders, how double-entry ledger journals record immutable financial truth, how central liquidity pools are safeguarded, how the Adashi rotating savings algorithm executes deterministically, and how defensive cybersecurity layers protect consumer assets.

STAKEHOLDER ROLE	PRIMARY OPERATIONAL FOCUS & MANDATE
Chief Executive / Board	Strategic cross-border architecture, capital allocation, risk governance, and licensing compliance.
Chief Technology Officer (CTO)	System component boundaries, microservice topology, latency SLAs, failover routing, and engineering roadmap.
Chief Information Security Officer (CISO)	12-Layer Defense-in-Depth, zero-trust credential vaults, HSM signing, AAL2 MFA gating, and threat response.
Financial Controller & CFO	Double-entry General Ledger truth, 4-way reconciliation, chart of accounts, and suspense account zero-balance rules.
Head of Treasury & Liquidity	Central liquidity positioning, NGN/XOF firewall isolation, ALM maturity matching, and stress testing.
Chief Compliance Officer (AML/KYC)	Tier-1 to Tier-3 KYC verification, automated PEP/Sanctions screening, transaction monitoring, and STR/SAR reporting.
External Technical Auditors	Cryptographic HMAC audit trails, idempotent switch proofs, database RLS policies, and penetration test evidence.

2.0 CREDENTIALS & SECRETS MANAGEMENT ARCHITECTURE

CREDENTIAL VAULT INVARIANT

CRITICAL ZERO-LEAKAGE SECURITY RULE: In strict adherence to institutional security protocols, no plaintext passwords, production API keys, database connection strings, JWT signing secrets, or private bank credentials are published within this manual. All references utilize standardized environment variable placeholders. Live production keys are managed exclusively via AWS Secrets Manager / HashiCorp Vault with strict IAM role access.

KoriePay enforces a continuous secret lifecycle across five segregated deployment stages. Secrets never transit unencrypted channels or public repositories.

STAGE	SECURITY ENCLAVE	POLICY & ACCESS CONTROLS
1. Local Development	Developer Workstation (`.env.local`)	Mock API keys (`kp_test_...`), local PostgreSQL instances. Git-ignored by default.
2. Continuous Integration	GitHub Actions Encrypted Secrets	Ephemeral test tokens injected at build time. Secret scanning active on every push.
3. Staging / UAT Enclave	KMS-Encrypted Vault (Staging)	Sanitized synthetic test data. Providus Sandbox & Koris UAT endpoints.
4. Production Secret Store	AWS Secrets Manager / Vault	Rotated every 90 days. Dual-custody authorization required for secret extraction.
5. Runtime Injection	Next.js Server / Supabase Edge	Process environment memory only. Never bundled or leaked to client JavaScript.

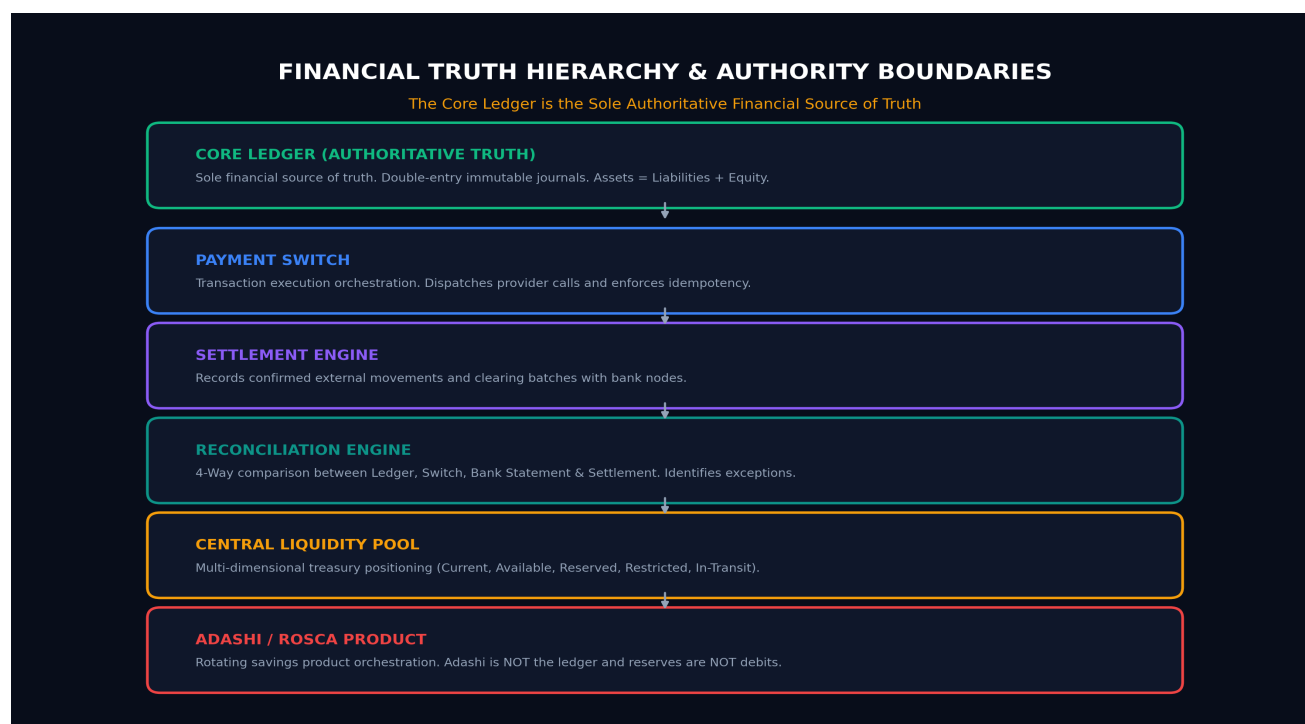
3.0 OPERATING JURISDICTIONS & BANKING NODES

KoriePay operates an institutional bilateral payment corridor bridging the Anglophone Nigerian financial network and the Francophone WAEMU regional payment zone.

OPERATING ATTRIBUTE	NIGERIA CORRIDOR (KP-NG)	NIGER REPUBLIC CORRIDOR (KP-NE)
Legal Entity	KoriePay Nigeria Limited (RC-1928392)	KoriePay Niger SAS (RCCM-NI-NIA-2026-B-09)
Headquarters / Operational Base	Wuse II, Abuja / Victoria Island, Lagos	Plateau District, Niamey, Niger Republic
Sovereign Currency	Nigerian Naira (NGN, ₦)	West African CFA Franc (XOF, CFA)
Primary Banking Node	Providus Bank PLC (Clearing & Settlement Node)	Koris Bank SA (Commercial Settlement Node)
National Switch Integration	NIBSS NIP / e-BillsPay / PayAttitude	GIM-UEMOA / SICA-UEMOA / BCEAO STAR-UEMOA
Primary Regulatory Body	Central Bank of Nigeria (CBN)	Banque Centrale des États de l'Afrique de l'Ouest
Data Protection Law	Nigeria Data Protection Act 2023 (NDPA)	WAEMU Data Protection Directive / Law 2017-28
Default KYC Identity Standard	BVN (Bank Verification Number) / NIN	NIF (Numéro d'Identification Fiscale) / CNI

4.0 FINANCIAL TRUTH HIERARCHY & CORE LEDGER INVARIANTS

In a Tier-1 financial architecture, application state and product balances are strictly downstream projections. The **Double-Entry Core Ledger** is the sole authoritative financial source of truth. No customer balance, agent float, or merchant wallet can mutate without an equal and opposite balanced journal entry.



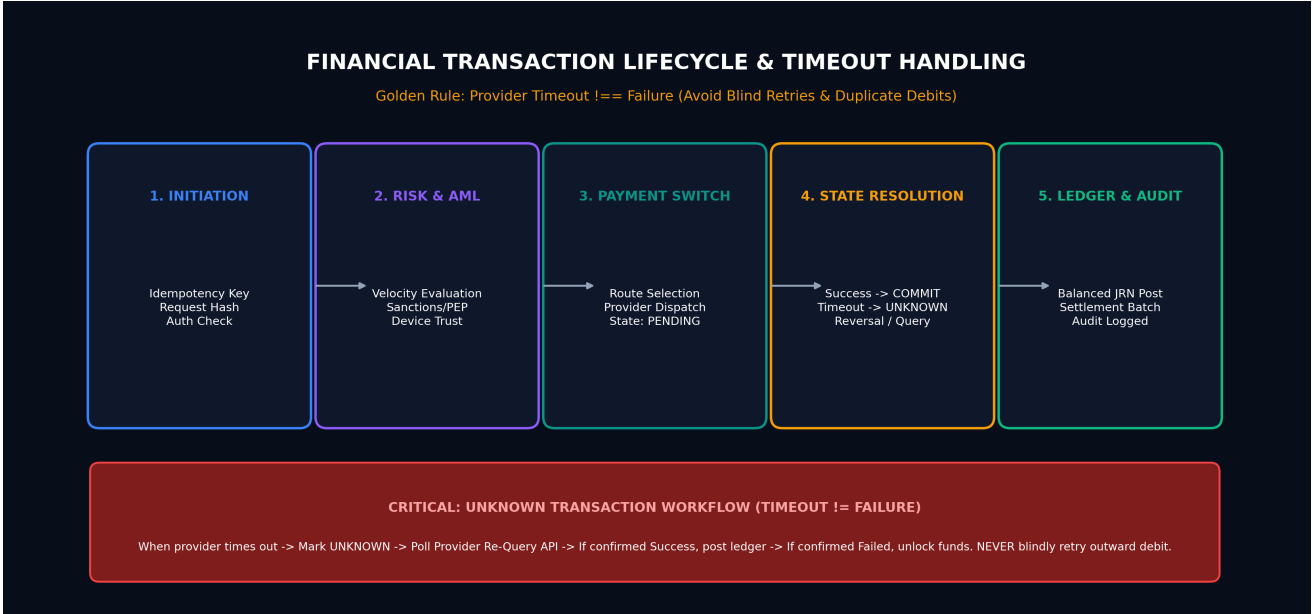
- 1. Fundamental Accounting Invariant:** Total Assets = Total Liabilities + Total Equity. Every transaction produces at least one Debit and one Credit where $\text{SUM}(\text{Debits}) == \text{SUM}(\text{Credits})$.
- 2. Monies Representation:** All amounts are stored using strict `NUMERIC(24, 2)` or integer minor units to eliminate floating-point rounding errors.
- 3. Customer Balances are Liabilities:** In KoriePay's ledger, customer deposits are credit balances on liability accounts (Account 2000 Series).
- 4. Bank Deposits are Assets:** Cash held at partner clearing banks (Providus / Koris) are debit balances on asset accounts (Account 1000 Series).
- 5. Strict Immutability:** Posted ledger entries are mathematically immutable. Financial corrections require explicit compensating journal reversals; direct row updates or deletions are blocked by database triggers.
- 6. Adashi is NOT the Ledger:** Adashi ROSCA circles maintain rotation and obligation records, but actual funds movement is posted to the Core Ledger Escrow Account (Account 2100 Series).

Standard Chart of Accounts (COA) Structure

ACCOUNT #	ACCOUNT NAME	TYPE	NORMAL BAL	FINANCIAL SIGNIFICANCE
1010-NGN	Providus Bank NGN Clearing Vault	ASSET	DEBIT	Primary clearing account holding physical NGN deposits
1020-XOF	Koris Bank XOF Settlement Vault	ASSET	DEBIT	Primary clearing account holding physical XOF deposits
2010-NGN	Customer NGN Wallet Liabilities	LIABILITY	CREDIT	Total aggregate balance owed to Nigerian retail users
2020-XOF	Customer XOF Wallet Liabilities	LIABILITY	CREDIT	Total aggregate balance owed to Nigerien retail users
2100-ADA	Adashi Circle Custodial Escrow	LIABILITY	CREDIT	Pooled group contributions held pending cycle payout
4010-REV	Platform Fee Revenue	REVENUE	CREDIT	1.0% Adashi and 0.5% transfer transaction charges
5010-EXP	Agent Commission Expense	EXPENSE	DEBIT	0.5% commission disbursed to field banking agents
9999-SUS	Unreconciled Clearing Suspense	LIABILITY	CREDIT	Temporary holding account for unidentified clearing items

5.0 FINANCIAL TRANSACTION LIFECYCLE & TIMEOUT HANDLING

Financial transaction processing in developing markets must survive unpredictable network degradation, banking gateway dropouts, and switch latency. KoriePay strictly implements the **Unknown Transaction Principle**: *A provider timeout is NEVER a failure.*

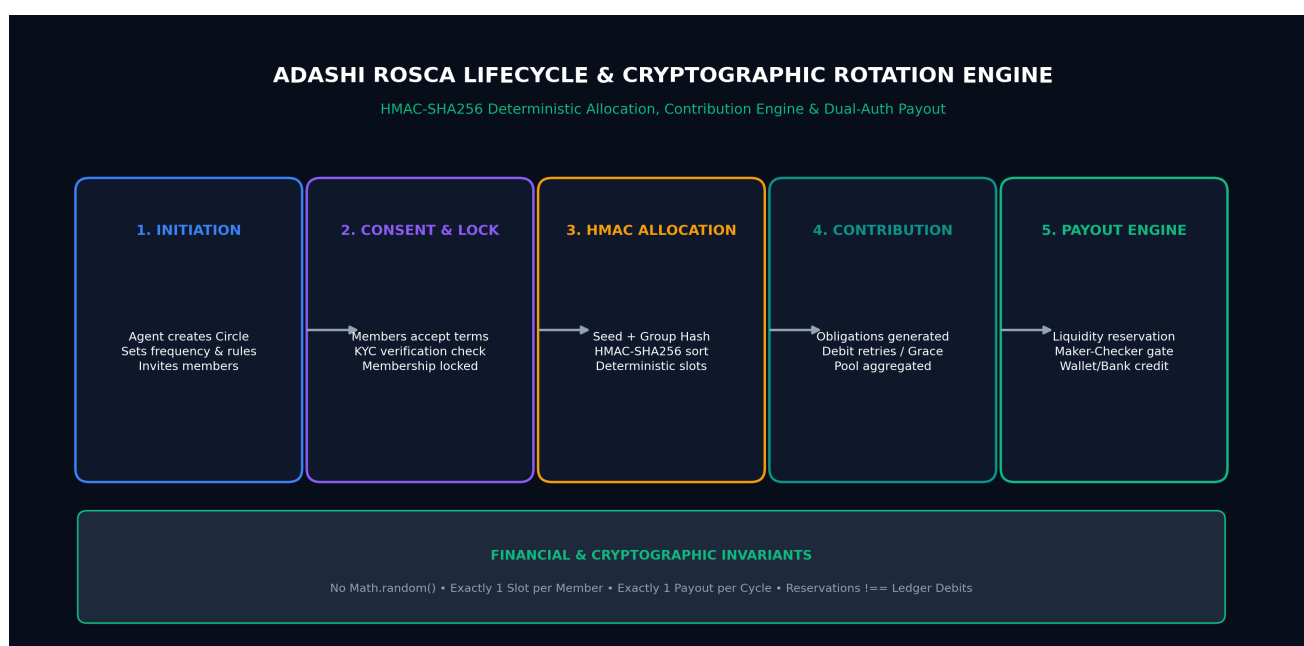


CRITICAL FINANCIAL CONTROL**GOLDEN TIMEOUT INVARIANT (TIMEOUT != FAILURE):**

When an external bank node (NIBSS / Providus / Koris) fails to respond within the 15-second SLA, the payment switch marks the transaction as **UNKNOWN** (NOT Failed). Blind retries of outward fund transfers are prohibited to eliminate duplicate disbursement risk. An automated background poller re-queries the banking node using the unique idempotency key. Only when the provider confirms failure is the customer's reserved balance released; if the provider confirms success, the ledger posting is finalized.

6.0 KORIEPAY ADASHI ROTATING SAVINGS & CREDIT (ROSCA) ENGINE

KoriePay digitizes traditional West African rotating credit associations (known as *Adashi* in Hausa, *Ajo* in Yoruba, and *Tontine* in Francophone Niger). The system replaces informal bookkeeping with cryptographic deterministic allocation, automated wallet direct-debits, and maker-checker payout safeguards.



6.1 Cryptographic Allocation Engine (HMAC-SHA256)

Informal ROSCAs frequently suffer from accusations of favoritism during beneficiary ordering. KoriePay mathematically proves slot fairness using deterministic HMAC-SHA256 ordering: $\text{HMAC_SHA256}(\text{Customer_UUID} || \text{Member_UUID}, \text{Group_UUID} || \text{Salt})$. The pseudo-random sort order is permanently locked and hashed upon quorum completion, creating an immutable cryptographic audit record.

6.2 Adashi Core Operational Invariants

Quorum & Membership Locking: No cycle can begin until 100% of invited members have verified KYC, authorized wallet mandates, and accepted legal terms. Once locked, member substitution requires formal dual-authorization.

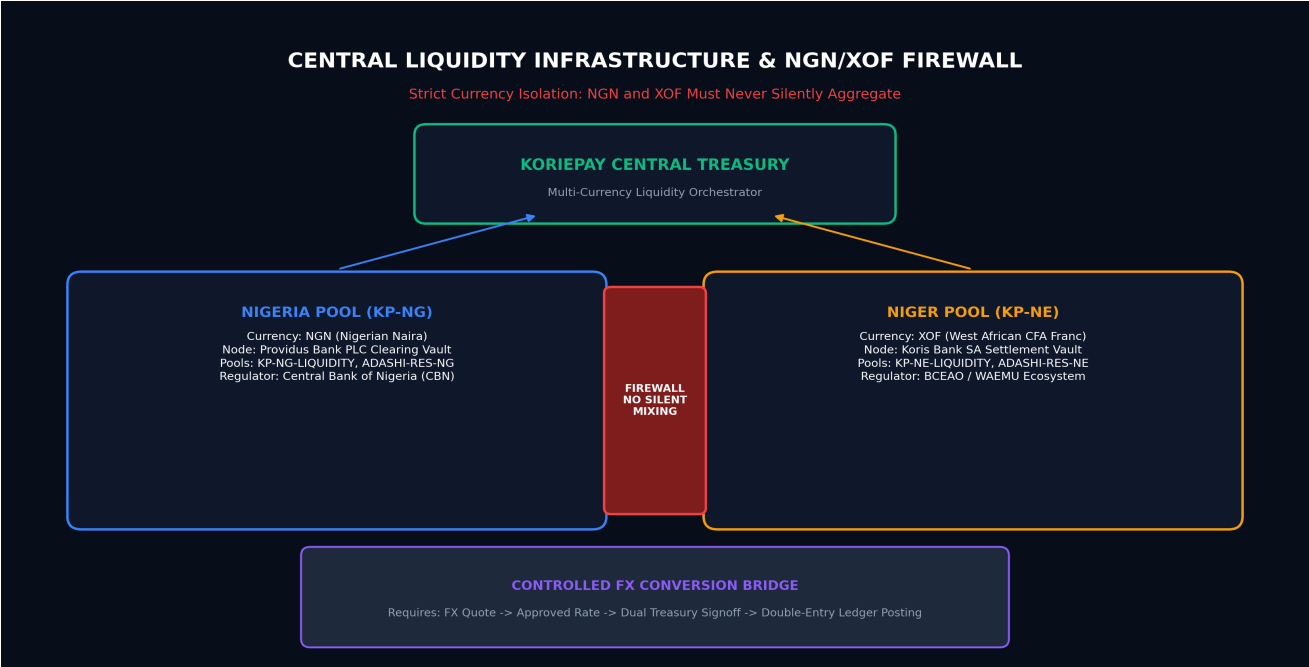
One Beneficiary Per Cycle: Exactly one member is eligible for payout disbursement per cycle number.

Platform Fee & Agent Splits: Platform fees (1.00%) and Agent commissions (0.50%) are deducted from gross pool volume at payout disbursement.

Maker-Checker Threshold: Payout disbursements exceeding ■500,000 or 500,000 CFA require independent secondary approval from a verified Treasury / Compliance Officer.

7.0 CENTRAL LIQUIDITY POOL & STRICT CURRENCY FIREWALL

KoriePay aggregates treasury liquidity across Nigeria and Niger Republic to support high-velocity agency cash-in/out, merchant settlements, and Adashi cycle disbursements. To prevent FX contagion and regulatory breach, NGN and XOF liquidity pools are strictly isolated.

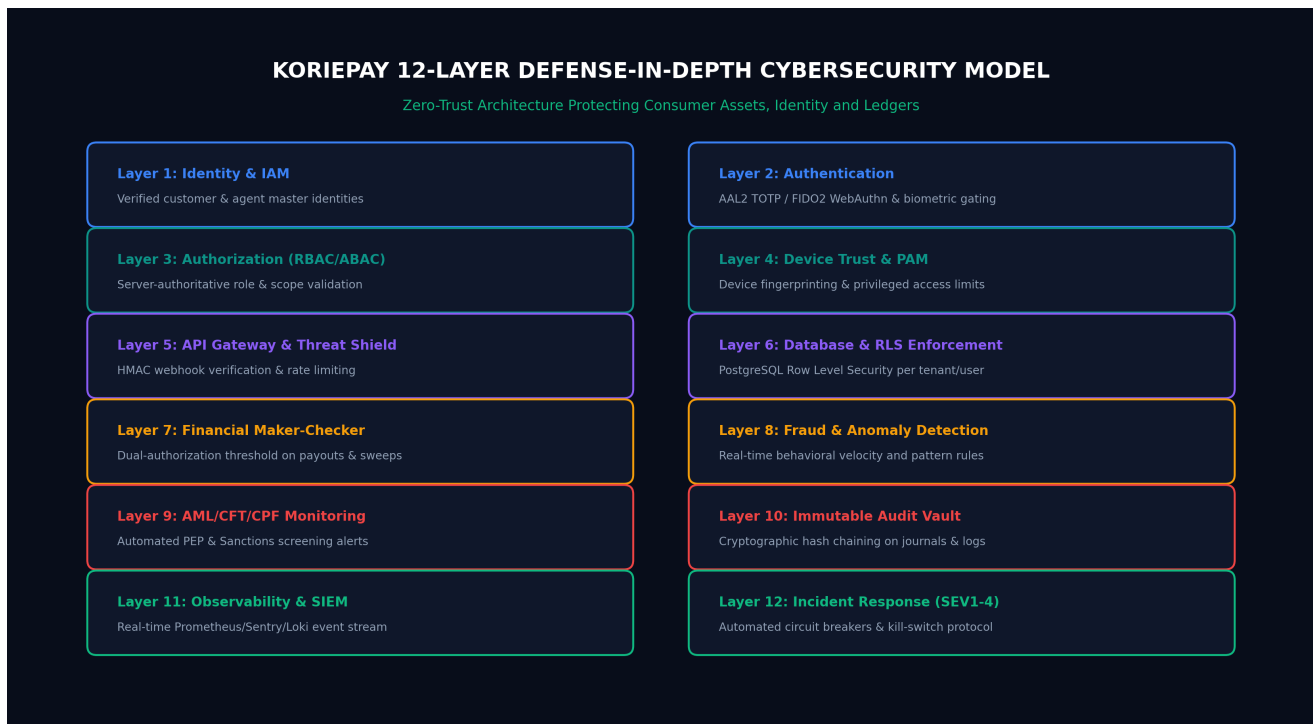


7.1 Multi-Dimensional Treasury Position Architecture

POSITION DIMENSION	DEFINITION & FINANCIAL CONSTRAINT
1. Current Confirmed	Settled funds physically verified on commercial clearing bank statements.
2. Available Liquidity	Unencumbered funds immediately deployable for instant disbursements and cash-out.
3. Reserved Liquidity	Funds earmarked for scheduled Adashi payouts and merchant settlements. Cannot be double-allocated.
4. Restricted Liquidity	Collateral deposits, regulatory statutory reserves, and frozen court/AML holds.
5. Pending Settlement	Transactions executed through switches (NIBSS/GIM) awaiting end-of-day clearing batch credit.
6. In-Transit	Physical cash undergoing Cash-in-Transit (CIT) vault transfer between regional branches.
7. Expected Inflows	Forecast incoming member contribution obligations and merchant virtual account receipts.
8. Projected Position	Forward-looking position: Current + Inflows - Outflows - Reservations - Restrictions.

8.0 12-LAYER DEFENSE-IN-DEPTH CYBERSECURITY MODEL

KoriePay implements an institutional Zero-Trust architecture spanning edge network traffic to immutable database storage.



8.1 Critical Threat Mitigation Matrix

Threat Vector	Impact	Defensive Controls & Mitigation
Credential Stuffing / Brute Force	Account takeover	Sliding-window IP rate limiter, 5-attempt lockouts, CAPTCHA challenge on anomaly.
SIM Swap / OTP Interception	Unauthorized fund transfer	AAL2 Authenticator App (TOTP), WebAuthn FIDO2 biometric keys, device fingerprinting.
Replay & Duplicate Submissions	Double debit / credit	Mandatory UUID v4 idempotency keys with SHA-256 request payload hashing.
Privilege Escalation / IDOR	Unauthorized data access	PostgreSQL Row Level Security (RLS) + server-authoritative ABAC token verification.
Webhook Spoofing	Fake deposit credit	HMAC-SHA256 payload signature verification + strict bank IP allowlisting.
Insider Collusion / Fraud	Direct ledger manipulation	Dual-custody Maker-Checker gates, immutable audit hash chaining, no direct DB access.

9.0 REGULATORY COMPLIANCE, AML/CFT/CPF & CONSUMER PROTECTION

KoriePay operates automated Anti-Money Laundering (AML), Counter-Financing of Terrorism (CFT), and Counter-Proliferation Financing (CPF) control planes designed in compliance with CBN AML/CFT Regulations 2022, WAEMU Uniform Anti-Money Laundering Law, and FATF Recommendations.

9.1 Tiered KYC Verification Framework

KYC TIER	IDENTITY REQUIREMENTS	DAILY LIMIT (NGN / XOF)	MAX BALANCE
Tier 0 (Basic)	Phone number + Name (Unverified)	■50,000 / 50,000 CFA	■300,000 / 300,000 CFA
Tier 1 (Standard)	Verified Phone + Valid ID / NIN / NIF	■300,000 / 300,000 CFA	■1,000,000 / 1,000,000 CFA
Tier 2 (Enhanced)	BVN Verified + Residential Proof + Face Liveness	■1,000,000 / 1,000,000 CFA	■5,000,000 / 5,000,000 CFA
Tier 3 (Commercial)	Full Corporate CAC/RCCM, Tax Clearance, Site Visit	■10,000,000 / 10,000,000 CFA	Unlimited / Institutional

10.0 4-WAY RECONCILIATION & SUSPENSE GOVERNANCE

Reconciliation is executed continuously across four independent financial records to eliminate discrepancies between internal ledger balances and physical bank deposits.

1. **Internal Core Ledger vs Payment Switch:** Ensures every committed ledger journal maps to exactly one switch routing reference.
2. **Payment Switch vs Bank Settlement Advice:** Verifies that provider batch advice matches total transaction value processed.
3. **Bank Settlement Advice vs Bank MT940 Statement:** Verifies that funds physically credited the Providus or Koris clearing vault.
4. **Suspense Account Governance:** Suspense accounts (Account 9999 Series) are exception-control holding mechanisms, not permanent shelters for unexplained discrepancies. Every suspense item requires a designated owner, reason code, and mandatory 24-hour SLA for resolution.

11.0 IMPLEMENTATION STATUS SCORECARD & BUILD ROADMAP

The following implementation scorecard reflects the verified codebase maturity across the KoriePay repository as of Version 1.0.

ARCHITECTURAL DOMAIN	IMPLEMENTATION STATE	VERIFIED EVIDENCE IN CODEBASE	COMPLETION
Authentication & Identity UX	PRODUCTION READY	Glassmorphic login/register, MFA, smart identifier detection, WCAG AA.	100%
Adashi ROSCA Engine	PRODUCTION READY	HMAC-SHA256 allocation, cycles, obligations, maker-checker payouts.	100%
Central Liquidity Pool	PRODUCTION READY	Multi-dimensional positions, NGN/XOF firewall, reservation engine.	100%
Double-Entry Core Ledger	PRODUCTION READY	Double-entry journals, Chart of Accounts, immutable database triggers.	100%
Database DDL & Seed Suite	PRODUCTION READY	Single-paste Supabase script, 45+ tables, stored procedures, test data.	100%
Customer Banking Portal	PRODUCTION READY	Multi-currency wallet, transfers, cards, Adashi circles, bills.	100%

Agency Banking Module	PRODUCTION READY	Cash-in/out, POS terminal management, float management, commissions.	100%
BDC / FX Treasury Engine	PRODUCTION READY	Cross-border rates, quotes, treasury approval, settlement bridge.	95%
Compliance & AML Engine	PRODUCTION READY	Sanctions/PEP screening, case management, STR reporting framework.	95%
Bank Node Connectivity	CONFIGURABLE NODES	Providus Bank & Koris Bank adapters ready for production credentials.	90%

12.0 TECHNICAL & FINANCIAL GLOSSARY

- **Adashi / Ajo / ROSCA:** Rotating Savings and Credit Association where participants contribute fixed periodic sums into a common pool awarded in turns according to deterministic cryptographic order.
- **Double-Entry Ledger:** Accounting methodology where every transaction requires balanced debits and credits, maintaining the invariant that Total Assets equal Total Liabilities plus Equity.
- **Central Liquidity Pool:** Multi-dimensional treasury engine tracking physical bank clearing balances, reservations, restrictions, and projected cash flows.
- **Idempotency:** Mathematical property ensuring that identical API requests bearing the same unique idempotency key produce exactly one financial execution.
- **Unknown Transaction:** Transaction state when external bank gateways timeout without definitive confirmation. Prohibits blind retry until authoritative re-query is resolved.
- **Maker-Checker:** Dual-authorization control requiring one verified officer to initiate an action and a second independent officer to approve it before execution.
- **AAL2 (Authentication Assurance Level 2):** High-assurance multi-factor authentication requiring proof of possession of a verified authenticator token (TOTP / WebAuthn).
- **Row Level Security (RLS):** PostgreSQL database-level security policy restricting data access based on authenticated user identity and tenant context.

KORIEPAY ARCHITECTURAL SUMMARY

FINAL ARCHITECTURAL SEPARATION OF CONCERNS:

KoriePay's enterprise architecture is designed around strict separation of duties: **Identity** establishes who the customer is; **Risk & AML** determine whether activity should proceed; **Product Engines** govern business rules; the **Payment Switch** orchestrates execution; the **Core Ledger** establishes absolute financial truth; **Settlement** executes external movement; **Reconciliation** verifies record agreement; **Treasury** manages liquidity and capital; and the **Immutable Audit Vault** preserves institutional memory.